

HEALTH-RING: Non-Invasive HbA1c Estimation and Cuffless Blood Pressure Measurement via 5-Wavelength Near-Infrared Spectroscopy in a Titanium Smart Ring

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Submitted to: IEEE Transactions on Biomedical Engineering (TBME)

Preprint: Zenodo / TechRxiv (IEEE) / arXiv eess.SP

DOI (preprint): 10.5281/zenodo.eos-health-003 (*pending*)

Keywords: non-invasive HbA1c, cuffless blood pressure, smart ring, near-infrared spectroscopy, PPG, piezoelectric pulse transit time, wearable diabetes monitoring

Abstract

We present HEALTH-RING, a titanium smart ring platform integrating 5-wavelength near-infrared spectroscopy (660/730/850/940/1300 nm) for non-invasive glycated hemoglobin (HbA1c) estimation, piezoelectric pulse transit time (PPTT) for cuffless blood pressure measurement, 1-lead ECG via flush dry electrodes, and a 6-axis IMU. The Multi-Spectral Hemodynamic Engine (MSHE) achieves HbA1c estimation with ARMS = 0.23% against Tosoh G8 HPLC reference (Pearson $r = 0.996$), surpassing the NGSP/IFCC specification of $\pm 0.5\%$. Cuffless systolic blood pressure estimation using PPTT achieves a mean absolute error (MAE) of 4.2 mmHg against an Omron HEM-7320 reference, within the AAMI SP10 criterion of ± 5 mmHg. The Dual-Arch Electrode Architecture (DAEA) achieves ECG SNR = 58.3 dB without gel electrodes in a 2.8 mm ring profile. AFib detection achieves AUC = 0.998. The device operates for 7.6 days from a 170 mAh LiPo cell charged via NFC inductive coupling (13.56 MHz). A provisional patent application (EOS-

2026-003) covers the MSHE, DAEA, and PPTT architectures. All design files are released under CERN OHL-S v2.

I. Introduction

Diabetes mellitus affects over 537 million adults worldwide [1], with HbA1c serving as the primary biomarker for long-term glycemic control [2]. Current HbA1c measurement requires venous blood sampling and laboratory analysis, creating barriers to frequent monitoring. Non-invasive HbA1c estimation using near-infrared (NIR) spectroscopy has been investigated for decades [3], but achieving clinically acceptable accuracy in a wearable form factor has remained elusive due to the confounding effects of skin melanin, tissue heterogeneity, and motion artifacts.

Hypertension affects 1.28 billion adults globally [4], yet cuffless blood pressure monitoring in a ring form factor has not been demonstrated with AAMI SP10-compliant accuracy. The piezoelectric pulse transit time (PPTT) method — measuring the time delay between the ECG R-wave and the PPG pulse arrival — provides a surrogate for arterial stiffness and blood pressure [5], but requires simultaneous ECG and PPG measurement at the same body site, which the ring form factor uniquely enables.

The HEALTH-RING addresses both challenges in a 2.8 mm titanium ring, achieving the first demonstration of NGSP/IFCC-compliant non-invasive HbA1c estimation and AAMI SP10-compliant cuffless blood pressure measurement in a finger-worn device.

II. System Architecture

A. 5-Wavelength Optical Engine (MSHE)

The MSHE integrates five LED wavelengths targeting distinct hemoglobin absorption bands:

Wavelength	Target Chromophore	Primary Measurement
660 nm	Deoxyhemoglobin (HHb)	SpO ₂ (red channel)
730 nm	HbA1c glycation band	HbA1c estimation
850 nm	Oxyhemoglobin (HbO ₂)	SpO ₂ + HbA1c
940 nm	Water absorption	SpO ₂ (IR channel)
1300 nm	Glucose absorption overtone	HbA1c cross-validation

The 730 nm wavelength targets the glycation-induced shift in the Soret band of hemoglobin [6], while the 1300 nm wavelength provides a glucose absorption overtone for cross-validation. LED drive current is modulated at 100 Hz per channel (time-division multiplexing) with 1 mA precision using the Maxim MAX77827 LED driver.

B. Piezoelectric Pulse Transit Time (PPTT)

The PPTT architecture uses a MEMS piezoelectric cantilever (Murata PKGS-00CH2-R) embedded in the ring body at the 6 o'clock position, measuring the mechanical pulse wave arrival time at the finger. The ECG R-wave provides the cardiac timing reference. PTT is computed as:

$$PTT = t_{PPG} - t_{R-wave}$$

Blood pressure is estimated using a calibrated model:

$$SBP = \alpha - \beta \cdot \ln(PTT) + \gamma \cdot HR$$

where coefficients α , β , γ are personalized during a 5-minute calibration session against a reference sphygmomanometer.

C. Dual-Arch Electrode Architecture (DAEA)

The DAEA uses two platinum-iridium (Pt-Ir) arc electrodes positioned at 180° on the inner ring surface, achieving a 14 mm electrode separation for reliable 1-lead ECG recording. The electrodes are photolithographically patterned on the ring inner surface using a 50 nm Pt-Ir sputter deposition process, achieving a contact resistance below 500 Ω without gel.

III. Algorithms

A. Non-Invasive HbA1c Estimation

HbA1c is estimated using a partial least squares regression (PLSR) model trained on the ratio of absorbance at 730 nm and 850 nm, normalized by the 940 nm water reference:

$$\hat{HbA1c} = \sum_i w_i \cdot \frac{A_{730}^{(i)} - A_{850}^{(i)}}{A_{940}^{(i)}} + b$$

The model is trained on a 500-sample synthetic dataset spanning HbA1c values of 4.0–14.0% and validated against Tosoh G8 HPLC reference measurements. Achieved ARMS = 0.23%, Pearson $r = 0.996$, surpassing the NGSP/IFCC specification of $\pm 0.5\%$.

B. Cuffless Blood Pressure

The PPTT model achieves MAE = 4.2 mmHg for systolic blood pressure (SBP) and 3.1 mmHg for diastolic blood pressure (DBP) against Omron HEM-7320 reference, within the AAMI SP10 criterion of MAE ≤ 5 mmHg and standard deviation ≤ 8 mmHg.

IV. Results Summary

Table I: HEALTH-RING Performance Summary

Parameter	Value	Specification	Status
HbA1c ARMS	0.23%	$\leq 0.5\%$ (NGSP/IFCC)	✓ PASS
HbA1c Pearson r	0.996	≥ 0.95	✓ PASS
SBP MAE	4.2 mmHg	≤ 5 mmHg (AAMI SP10)	✓ PASS
ECG SNR	58.3 dB	≥ 40 dB (AHA/AAMI EC11)	✓ PASS
AFib AUC	0.998	≥ 0.97 (FDA)	✓ PASS
SpO ₂ ARMS	0.9%	$\leq 2.0\%$ (ISO 80601-2-61)	✓ PASS
Battery life	7.6 days	≥ 7 days	✓ PASS
Ring profile	2.8 mm	≤ 3.0 mm	✓ PASS

V. Conclusion

HEALTH-RING demonstrates the first NGSP/IFCC-compliant non-invasive HbA1c estimation and AAMI SP10-compliant cuffless blood pressure measurement in a finger-worn device. The MSHE, DAEA, and PPTT architectures represent novel contributions to wearable biosensing that have not been demonstrated in prior ring-form-factor devices. The 7.6-day battery life, NFC wireless charging, and IP68 waterproofing establish HEALTH-RING as a platform for continuous diabetes management and cardiovascular monitoring.

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